



Mihai Minoiu

Robotics Perception Engineer | Computer Vision & ROS2 | Python, C++, Linux & NVIDIA Jetson
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Portfolio: minoiu-mihai.github.io

Summary

Robotics Perception Engineer focused on real-time computer vision, visual tracking and Edge AI deployment for robotic and embedded systems.

Experienced in ROS2-based perception pipelines using Python, C++, OpenCV, ONNX, NCNN, TensorRT and NVIDIA Jetson, with a mechanical/aerospace background for HW/SW integration.

Core Skills

Robotics & Vision	Edge AI	Software Engineering	Mechanical & Systems Design
ROS2, OpenCV, Computer Vision, Visual Tracking, Segmentation, Object Detection	NVIDIA Jetson, PyTorch, ONNX, TensorRT, CUDA, NCNN	Python, C++, Linux, Git, Docker, YAML	CATIA V5, PLM, Airbus Methodology, HW/SW Integration

Professional Experience

Adatica Engineering	Madrid
Robotics & Perception Engineer	2025-Present
- Developed real-time ROS2 perception pipelines in Python and C++ for UAV and embedded robotic platforms, integrating detection, segmentation, visual tracking and reinitialization workflows. - Deployed AI inference on NVIDIA Jetson using PyTorch, ONNX, NCNN and TensorRT, optimizing low-latency execution. - Migrated CPU-based detection workflows to GPU-accelerated TensorRT inference, achieving up to 6x throughput improvement.	
Industrial Design Engineer / Mechanical Designer	2024-2025
- Executed CATIA V5 and PLM workflows under Airbus-based engineering methodology for aerospace and industrial mechanical design. - Designed and reviewed mechanical components with focus on structural integrity, manufacturability and hardware/software integration constraints.	
Advanced Mineral Processing	Madrid
Field Service Technician (Mining Division)	2023-2024
Troubleshoot and maintained heavy machinery in demanding industrial environments, developing practical understanding of failure modes, field reliability and high-pressure technical decision-making.	

Projects

Real-Time Visual Tracking Pipeline
Built a modular perception pipeline combining segmentation, fast tracking, rollback correction, confidence gating and reinitialization logic.
Selective Pesticide Application Robot
Developed an embedded vision-and-actuation prototype for selective pesticide application using Jetson-based perception and ESP32-controlled pumps.
MEMENTO Tracking Arbitration Layer
Designed a tracker-agnostic memory layer for validating bounding boxes, detecting drift and managing uncertain/lost tracking states.

Education

EDDM Engineering	Aerospace Design Engineering Bootcamp
Madrid • 2024-2025	
Intensive Aerospace Design Bootcamp based on Airbus internal methodology, specialized in structural engineering and PLM-driven development.	
Nebrija University	Bachelor's Degree in Industrial Design Engineering.
Madrid • 2018-2025	
Focused on engineering fundamentals including machine theory, circuit theory, strength of materials, technical design and design thinking.	

Publications

Aplicación Selectiva de Fitosanitarios Mediante Visión Artificial	10/2025
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Languages

Spanish	Romanian	English	German
Native	Native	B2	A1